CVE-2026-46242 Tier 2 Progress — Executive Summary

Project: bad-epoll-lab · **Branch:** tier2-android-port @ cc0dc7754 (verified on origin) · **Kernel:** linux-6.12.67, Android 14 GKI (commit 7e35917775b8) · **Date:** 2026-08-07

Verdict: The epoll UAF is real and reproducible with debugger assistance (hardware-watchpoint proven, VER-026), and msg_msg reclaim of the freed `struct eventpoll` in kmalloc-192 is reliable (VER-027). However, the race is **not naturally schedulable**: 0 / 102,740 unaided attempts (VER-034, VER-039). All four exploitation chains are structurally dead (21 dead ends); the only primitive is a fixed NULL write at offset 160, which EXP-016 shows is DoS-only on this configuration. **Recommendation:** 2-week hardware timebox (timing-widening on physical ARM64) with kill-criteria, else conclude DoS-only.

1. What was proved (RUNTIME evidence)

- VER-026 (EXP-015): two-threaded race — Thread A clears `f_ep`, Thread B bypasses `eventpoll_release_file` via the lockless fast path and frees `inner_epoll`; Thread A's `hlist_del_rcu` writes NULL at offset 160 of the freed object. Captured by hardware watchpoint.
- VER-027 (EXP-018/019): msg_msg with 144 B payload reclaims the freed slot with attacker control from byte 48.
- VER-020: `sizeof(struct eventpoll)=176` → kmalloc-192. VER-038: SysV IPC works on the target kernel (AND-001).

2. What was disproved (each with killing evidence)

- **Chain 0 — percpu_counter_dec crash: uses OUTER epoll, never the freed inner (VER-028).**
- **Chain 1 — dual-watch KASLR leak: single-epitem UAF and multi-epitem pointer write are mutually exclusive (VER-033; VER-029/030 retracted).**
- **Chain 2 — arbitrary decrement via fake user_struct: decrement always runs on root_user (VER-031/032).**
- **Chain 3 — full LPE: depends on Chains 1+2. Only remaining primitive: NULL @ offset 160 → EXP-016 audit shows crash-only/benign outcomes for all reachable kmalloc-192 structs (fib6_info, snd_timer_user, packet_fanout...).**
- **epitem same-cache reclaim (VER-016), struct file UAF (VER-018/025), snd_timer_user theories (EVO-005/VER-013).**

3. Why the natural race fails

- cond_resched() at eventpoll.c:888/903 is a NO-OP: dynamic_cond_resched static key is FALSE and TIF_NEED_RESCHED is never set on the pinned, idle CPU (NAT-002 correction, VER-035).
- __ep_remove has zero preemption points (disassembly-confirmed). The window is pure instruction timing: ~250–550 cycles (~125–275 ns @ 2 GHz).

- QEMU TCG does not model hardware cache-coherency/memory-bus timing. NAT-005 (92,740 iterations, isolcpus=1, nohz_full, 4 MB cache-eviction sweeper) reached a 1-cycle (~16 ns) best alignment error with 0 hits.

4. Where we are

Metric	Value
Experiments executed	19 (EXP-006..024, NAT-001/002/005, AND-001)
GDB-assisted UAF hits	~100% on demand
Natural race hits	0 / 102,740
Best timing alignment error	1 cycle (~16 ns)
Dead ends	21 (4 chains · 5 objects · 4 primitives · 4 sprays · 4 misconceptions)
Retracted claims (kept visible)	VER-010, VER-029, VER-030
Active verification entries	VER-009 .. VER-039
Android portability	AND-001 PASSED (SysV IPC); AND-002/003/004 planned

5. Decision options

Path	Description	Effort	Risk	Probability
A: Hardware timing-widening	Physical ARM64 device; false-sharing cache bouncing, slab contention, IPI/timer storms	2–4 weeks	Medium	Medium — only path with theoretical basis
B: Alternative race variant	Race where the freed object IS the ep parameter of __ep_remove	2–4 weeks	High	Low
C: Conclude DoS-only	Document all dead ends + statistical negatives; publish negative-result writeup	~1 week	Low	High confidence in conclusion

Hybrid recommendation: Path A for a strict 2-week timebox with kill-criteria (0 hits in ~1M iterations ⇒ Path C).

6. Reproducibility pointers

- Protocol: `tier2/docs/EXPERIMENT_PROTOCOL.md` (10 rules) · Runner: `tier2/docs/RUNNER_GUIDE.md` · Ledger: `tier2/docs/VERIFICATION_LEDGER.md`
- Dead ends: `tier2/docs/DEAD_ENDS_REGISTER.md` · Assumptions: `tier2/docs/ASSUMPTIONS_REGISTER.md`
- All raw evidence: `tier2/evidence/` · Harnesses/GDB scripts: `tier2/scripts/`
- Build & run: `DEBUG=1 ./scripts/run_qemu.sh`, `gdb -batch -q -x scripts/<S>.py android/artifacts/vmlinux`